

Kamran Amini

Senior Systems & Solutions Architect

Independent Consultant

- ✓ CTO Advisory & Strategic Decision Making
- ✓ Systems & Solutions Architecture
- ✓ Blockchain & AI Adoption
- ✓ Data & Network Security

Author of Books & Articles on Software Engineering

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SUMMARY

With 15+ years of experience across software development and systems architecture, I have worked in startups, large corporations, and consulting engagements as an engineer, Tech Lead, CTO, board member, and founder, giving me first-hand insight into the technical, financial, and organizational dynamics of software firms at every scale. I combine hands-on engineering with architectural leadership, coding critical components myself so my decisions stay informed by implementation reality. My core expertise lies in systems-level languages (C, C++, Golang, Rust), plus DevOps on Kubernetes and AWS/GCP. In blockchain and fintech, I architected tokenized-securities (RWA) infrastructure (on-chain exchange and auction engines with atomic settlement, a central securities depository, brokerage and custody) plus ERC-20/ERC-721 token and NFT platforms. I now build AI platforms end to end (self-hosted LLMs, vision and diffusion models, LoRA training) and work daily with AI coding agents (Claude, Codex, Gemini). My background is in data and network security, and I treat security and privacy as first-class citizens in my architecture and consulting. I bring this combined perspective to the systems architecture, CTO-level strategy, and blockchain/AI consulting I provide today.

WORK EXPERIENCE

DeKrypt B.V.

Founder, CEO & CTO, Principal Architect & Consultant

March 2024 - Present (alongside employment, with consent)

Amsterdam, The Netherlands

- **Consultancy:** Delivering all part-time architecture engagements listed in this CV through the firm (freelance before its founding).
- **SoulForge (own startup):** AI persona platform: self-hosted LLM/VLM inference serving, RAG memory on a vector database, FLUX image generation, per-persona LoRA training, face-identity conditioning, elastic GPU nodes.
- **FamBuzz & Bo2Bo2 (own startups):** A private group social app and a mobile fitness game: Golang backends on Kubernetes via Terraform and CI/CD, Android/iOS clients.
- **AI-Driven Development:** Delivering with AI coding agents (Claude, Codex, Gemini), keeping accountability with engineers.
- **FinTech & Distributed Ledger:** Architecting RWA exchange, settlement, MPC custody, and DeFi treasury systems for Vero Labs, Assetera, and Metapolitan.
- **Writing & Research:** Authoring **The Software Ecology** (to be published), alongside essays and academic work for peer review.

Vero Labs Inc.

Senior Blockchain Architect (Part-time consultancy)

April 2024 - September 2026

New York, United States

↳ Tokenise Ltd

Senior Blockchain Architect (Part-time consultancy)

London, United Kingdom

↳ VeroPulse GmbH

Chief Software Architect & Tech Lead (Part-time)

Vienna, Austria - led a team of 6

- **Agora Stock Exchange Engine:** Architected the RWA exchange's on-chain (DeFi) orderbook and settlement engine, with atomic settlement finality between the orderbook and treasury contracts.
- **Tokenise CSD:** Designed the Central Securities Depository for on-chain issuance, MPC-wallet custody, and settlement of tokenized securities (RWA).
- **Tokenise Broker:** Architected the brokerage platform connecting investors to the Agora Engine for order routing and execution.
- **Vero Android App:** Directed the architecture of the primary and secondary market mobile client, integrated end-to-end with the Agora Engine.
- **Platform & Infrastructure:** Designed the off-chain Golang microservices, deployed on Kubernetes (AWS EKS) via Terraform and CI/CD for scalable, highly-available trading workloads.
- **AI-Assisted Engineering:** Introduced AI-assisted coding workflows (LLM copilots, automated code review) across the teams to accelerate delivery and improve code quality.
- Golang - Kubernetes (EKS) - AWS - Terraform - Solidity - EVM chains - PostgreSQL - Docker - gRPC

Assetera GmbH

Senior Blockchain Architect & Tech Lead (Part-time)

April 2024 - March 2026

Vienna, Austria

- **Agora Auction Room:** Architected the on-chain (DeFi) auction mechanics for RWA tokens, with atomic settlement finality enforced against the on-chain treasury.
- **Tokenization as a Service:** Architected the on-chain framework for tokenizing real-world assets (RWA): issuance, MPC-wallet custody, and transfer.
- **Platform & Infrastructure:** Designed the off-chain Golang services, deployed on Kubernetes (AWS EKS).
- Golang - Solidity - EVM chains - Kubernetes (EKS) - AWS

Metapolitan Ltd.

Chief Blockchain Architect (Part-time consultancy)

January 2024 - March 2026

London, United Kingdom

- **Distributed Treasury System:** Architected a multi-signature and MPC-wallet on-chain treasury platform for RWA and DeFi clients, enabling secure custody, automated settlement, and policy-driven fund controls across multiple blockchain networks.
- Solidity - Gnosis Safe (multi-sig) - MPC wallets - EVM chains - web3.js - Hardhat - Node.js

Digital First GmbH

Senior Blockchain Architect (Part-time consultancy)

December 2021 - March 2024

Vienna, Austria

- **TheKiss NFT Tokenization Platform:** Designed the NFT minting architecture and smart contract framework underpinning the platform's digital collectibles.
- **ARTEO Token:** Architected the ERC-20 token economics and smart contract suite for the art-backed digital asset.
- **QLINDO Token:** Led smart contract architecture and security review for the project's token launch.
- **Hergé NFTs:** Architected the ERC-721 minting and royalty framework for the licensed NFT collection.
- Solidity - EVM chains - ERC-20 - ERC-721 - Hardhat - OpenZeppelin

ActiveVideo B.V. (April 2019 - February 2023)

↳ **Commscope Inc.** (March 2023 - March 2024)

↳ **ARRIS B.V.** (April 2024 - December 2025)

↳ **Vistance Networks B.V.** (January 2026 - Present)

Senior Software Engineer

Hilversum, The Netherlands - one company, many reorganizations

- **CloudTV (Video Stitching Engine):** Low-latency cloud renderer/streamer delivering UI and audio/video to set-top boxes.

- **Media Pipelines:** Built audio/video pipelines on MPEG-TS and H.264 with ffmpeg, including CEA-608/708 closed captions carried in SEI.
- **AppCloud:** Containerized Android streaming platform with a Kubernetes (AWS EKS) backend, provisioned with Terraform and shipped through CI/CD.
- **Multiview Engine:** High-throughput cloud streamer broadcasting multiple channels at once to set-top boxes and TVs.
- **Tech Council & Security Guild:** Prioritized tech debt; drove threat modeling and secure SDLC across teams.
- C++17 (g++, clang) - Golang - Java and Kotlin - Python - AWS - Kubernetes (EKS) - S3 - EFS - EBS - Ceph - Envoy

TomTom B.V.

Senior Software Engineer

September 2017 - March 2019

Amsterdam, The Netherlands

- Expert Software Consultant from Modis B.V.
- NavKit Free Text Search Engine, NavKit Worker REST Services, POI Info Services
- C++11 (g++, clang) - Java SE - JUnit - Spring MVC - JAXB - Spring REST - gdb - Valgrind - Clang ASAN & TSAN - STL & Boost - Google Test - Protobuf - Thrift - Python - CI/CD (Jenkins) - Docker - JIRA - Fisheye - Confluence - Bitbucket Stash - AWS/Azure based cloud

Jeppesen Systems (a Boeing subsidiary)

Senior Software Engineer

February 2017 - August 2017

Göteborg, Sweden

- Manpower Application
- C++11 (g++) - Python - C#.NET - gdb - Valgrind (callgrind, massif) - PDB - STL & Boost - Google Test - XMLRPC - Jenkins - JIRA - Fisheye - Confluence

Adro Co.

Senior Software Architect, Principal C++ Engineer

June 2016 - January 2017

Tehran, Iran

- Low-latency, high-throughput Realtime Bidding (RTB) Engine
- C++ (g++) - Python - gdb - Valgrind - STL & Boost - libuv - Google Test - Protobuf - Jenkins - Travis CI - Docker - JIRA - Fisheye

Naeem Co.

Chief Technical Officer, Chief Software Architect

April 2015 - October 2016

Tehran, Iran

- **Projects:** ePassport Issuance System - Face Recognition System, Biometric Identification System
- **Systems & Security:** Performance-critical C/C++ services; threat modeling and secure SDLC for PKI systems
- **Cryptography:** OpenSSL - PKCS (1, 7, 10, 11, 12) - ICAO MRTD standards - HSM - Key Management - PKCS11 Tokens
- **Stack:** C (gcc) - C++ (g++) - Java SE - Java EE - Spring MVC - Wildfly - Tomcat - Python - gdb - Valgrind - STL & Boost - wxWidgets - Qt - Google Test - Jenkins - Docker - vSphere - Proxmox - Gluster - Cassandra - Redmine - GitLab

Sharif CERT Center

Software Engineer, CERT Team Member

December 2011 - June 2013

Tehran, Iran

- Member of the Computer Emergency Response Team (CERT): incident response, threat analysis, and building the tooling used to detect and contain network threats.
- Botnet Detection System - Sinkhole - DDNS as a FreeBSD module
- C++ (g++) - Java SE - Perl - gdb - JConsole - Valgrind - STL & Boost - Swing - Redmine

TECHNICAL SKILLS

- **Languages & Architecture:** C++ (14/17/20), C, Golang, Rust, Python, Java, JS - Distributed systems (CAP, replication, leader election, CQRS, event sourcing, message queues, streaming, async/event loops) - Low-latency, high-throughput design
- **Cloud & DevOps:** Kubernetes (kubect1, Helm, Kustomize, PSA) - AWS & GCP (EKS/GKE, EC2/GCE, S3, VPC, IAM) - Terraform, CI/CD - Docker - PostgreSQL, MySQL, MongoDB, Redis
- **Blockchain:** Solidity (EVM), Solana, web3.js/ethers.js, Hardhat - Distributed ledgers, smart contract architecture, RWA, DeFi, MPC wallets
- **Security & Cryptography:** OpenSSL, PKI, symmetric/asymmetric algos, hashing/MAC - Protocols & standards (gRPC, Protobuf, REST, PKCS, RFCs) - Networking (TCP/IP, Wireshark, tcpdump) - Threat modeling, secure SDLC, incident response - CISSP (in progress)
- **AI & LLMs:** Self-hosted LLM/VLM & diffusion pipelines (Qwen, FLUX, Diffusers, Ollama) - LoRA training, identity conditioning (PuLID/IP-Adapter, face embeddings), inpainting, prompt compilation, RAG and agent memory (vector databases), elastic GPU inference serving - Classical ML & computer vision (TensorFlow, OpenCV, CUDA) - Agentic coding & review with Claude, Codex, and Gemini
- **Media & Streaming:** Audio/video pipelines - MPEG-TS, H.264, ffmpeg - Closed captions (CEA-608/708 in SEI)
- **Embedded & Kernel Development:** Linux kernel modules & system calls - FreeBSD kernel modules - Low-level C/C++ (memory, concurrency, IPC)

PUBLICATIONS & OPEN SOURCE PROJECTS

- **The Software Ecology** (to be published, DeKrypt B.V.): Examines the software industry as a socio-technical ecology, where companies, teams, and regulators interact through human behavior, power, and organizational dynamics as much as technology.
- **Extreme C:** Book published October 2019 by Packt Publication.
- **Linux Two-Step Authentication with Rust:** liveProject for Manning Publications, with its accompanying ESS backend and client projects in Rust.
- **Other Rust projects:** Domain Catcher Engine and the Outlook .msg parser.

LANGUAGES

Persian (**Native**), English (**Fluent**), German (**A1**), Dutch (**B1**), French (**A1**)

EDUCATION

- **Bachelor of Science in Physics (Astronomy and Planetary Science)**, The Open University, started 2017 (3 years completed, on leave)
- **Bachelor of Software Engineering**, University of Isfahan (Isfahan, Iran), 2004 - 2008

PERSONAL INTERESTS

- **Physics and Astronomy:** Cosmology, Astronomy, Quantum Physics, Quantum Mechanics, Particle Physics, Special and General Relativity, Theory of Everything, Photonics.
- **Mathematics:** Linear and Abstract Algebra, Probability Theory, Information Theory
- **Cryptography:** Information Security, Network Security, Security Algorithms, Quantum Information Theory, Quantum Key Distribution, Quantum Cryptography
- **Philosophy:** Western Philosophy, Sociology, Economics, Politics and Humanities
- **Telecommunications:** Modulation, Multiplexing, Photonics, optical fibers, IEEE 802.3, Coding Theory